01/19/2022

Tuesday, January 18, 2022 4:00 PM

M168_Notes_1.3



### *1.3 Linear Functions and Mathematical Models*

A linear model is a model that is a linear function,

$$f(x) = y = mx + b$$

$x$ is the independent variable (horizontal axis)
$y$ the dependent variable (vertical axis).
The domain is the allowed values for the independent variable.

Depreciation

- An item has an initial value and a final (scrap) value.
- It is assumed that the value decreases linearly with time

*Example*: A car is purchased for $15,000 and is kept for 5 years and at the end of the 5 years it is worth $5000. Find an expression for the value of the car as a function of time and graph it. What is the car worth after 4 years? What is the rate of depreciation of the car?

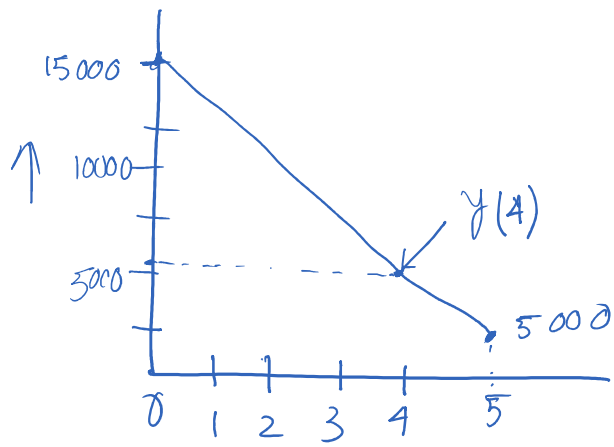
$y = mx + b$

$x = 0, \quad y = 15000, \quad x = 5, \quad y = 5000$

$m = \dfrac{15000 - 5000}{0 - 5} = -2000, \quad b = 15000$

$\boxed{y = -2000x + 15000}$

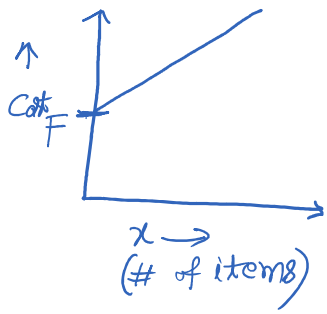
$y(4) = -2000 \cdot 4 + 15000 = \boxed{7000}$




Rate of depreciation = $2000/yr

Cost

- The cost of producing $x$ items is written $\boxed{C(x) = cx + F}$
- The fixed costs, $F$, are those costs that are independent of the number of items produced.
- The variable costs are those costs that vary as the number of items produced. The slope $c$ is the cost to make one more item.

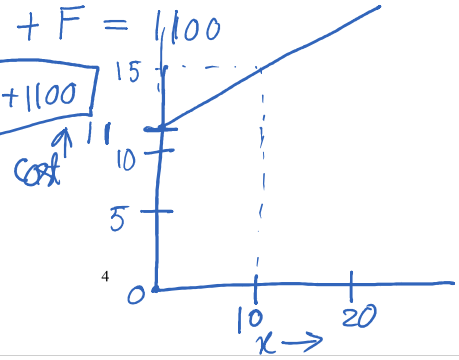



*Example*: A company makes heaters. They find that the cost to make 10 heaters is \$1500 and the cost to make 20 heaters is \$1900. Find the cost equation and graph it.

$(10, 1500)$, $(20, 1900)$, $c = \frac{1900-1500}{20-10} = 40$

$1500 = 40 \cdot 10 + F = 1100$

$\boxed{C(x) = 40 \cdot x + 1100}$



Revenue

If $x$ items are sold for $s$ dollars each, the money brought in, or revenue, from the sale of these items is

$$R(x) = sx$$

*Example*: Example - What is the revenue from selling heaters if the heaters sell for $50 each?

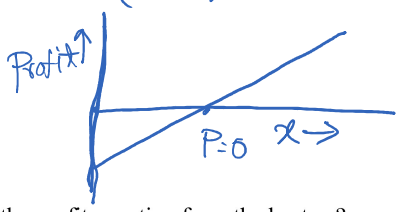
$$R(x) = 50 \cdot x$$

Profit

The profit made from selling $x$ items is $P(x)$ and it is the difference between the revenue (money in) and the cost (money out).

$$P(x) = R(x) - C(x)$$

$$P(x) = s \cdot x - (Cx + F) = (s - C)x - F$$




*Example:* What is the profit equation from the heaters?

$$P(x) = (50 - 40)x - 1100 = 10x - 1100$$

$x = \#$ items



<u>Demand</u>

$p$ is the price of an item. The demand is $D(x) = p$. This models the relationship between the price and the number of items purchased by the consumer.

*Example:* Say a store finds that it can sell 10 trench coats when the price is \$180 each and that is sells 50 trench coats when the price is \$100 each. Find the demand equation.

$(10, 180), (50, 100)$

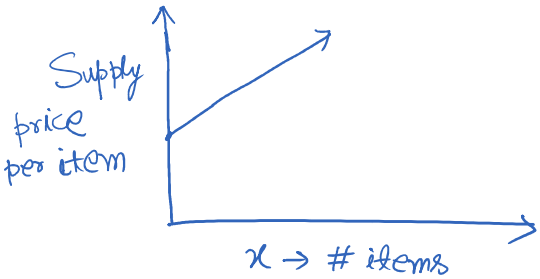
$y = mx + b, \quad m = \frac{180 - 100}{10 - 50} = -\frac{80}{40} = -2$

$180 = -2 \cdot 10 + b \Rightarrow b = 200$

$y = -2x + 200 = D(x)$

<u>Supply</u>

The supply function $S(x) = p$ models the relationship between the price of an item and how many are supplied to the market.




*Example:* A company manufactures trench coats. The company is not willing to sell trench coats unless it can get \$60 each. It will supply 10 trench coats if it can get \$80 each. Find the supply equation.

$(x_1, y_1) = (0, 60)$, $(10, 80) = (x_2, y_2)$

$y = mx + b$, $m = \dfrac{y_1 - y_2}{x_1 - x_2} = \dfrac{60 - 80}{0 - 10} = \dfrac{20}{10} = 2$

$60 = 2 \cdot 0 + b \Rightarrow b = 60$

$y = 2x + 60 = S(x)$